Knowledge Representation for Model Fair-Lending Validation

B.1 Submission Form

Field	Entry
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Industry / Domain	Banking and financial services / fair-lending validation
Organisation Type	Financial institution
[Instructor prompt]	knowledge representation using propositional logic, predicate logic and a knowledge base.

B.2 Anchor (1 mark)

Workplace artifact

I work as a risk analyst. The artifact is the checklist and review record I use to decide whether a model needs fair-lending review, which risk category applies, and which tests and approvals must be on file before the review can be closed. I refer to it as the Model Fair-Lending Validation Decision Checklist; this is a descriptive name, not the official document title.

Why it matters

Models influence who is offered credit, on what terms and how customers are treated. A wrong category, a missed test or an unexplained exception can leave a discriminatory outcome undetected or lead to inconsistent reviews of similar models. Fair-lending review looks for discriminatory treatment or outcomes under the Equal Credit Opportunity Act (ECOA) and, for residential real-estate credit, the Fair Housing Act (FHA) (OCC: Fair Lending).

How the process flows

1. **Submit:** The model owner provides the business purpose, intended use, methodology, inputs, outputs and documentation.
2. **Classify and assess:** The reviewer decides whether the tool is a model, records consumer impact and materiality, and proposes a risk category.
3. **Validate:** The analyst checks conceptual soundness, data suitability, implementation and performance, the core elements of the SR 11-7 validation framework (SR 11-7 attachment).
4. **Test for fair lending:** The analyst selects tests under internal policy, such as correlation screening of input features against protected-class contrasts, text controls, disparate impact analysis (DIA), an alternative approach or vendor assurance, depending on the use case and the data available.
5. **Record and decide:** Conclusions, legal advice, remediation items and the governance decision are documented.

Who is involved and where the facts come from

The model owner supplies use-case and input evidence. As the risk analyst I assess materiality, record the risk category, select and interpret tests, and document conclusions. The legal reviewer gives legal guidance, resolves unclear materiality and reviews vendor affirmations. A governance reviewer takes the final decision. Facts come from the model inventory, development documents, data dictionaries, validation datasets, monitoring reports and the recorded approvals.

Why a rule-based knowledge base fit

The checklist is a set of repeatable if-then decisions. A knowledge base can apply them the same way every time, show which facts support each conclusion, and send missing or ambiguous evidence to a person instead of guessing.

B.3 Apply (2 marks)

B.3.1 Cases used

Seven synthetic assessments exercise different branches of the rules.

Case	Use	What it exercises
a01 Underwrite_A	AL	Application-lending model; High category with a feasible DIA; used for the override query.
a02 Marketing_B	DM	Material direct marketing; Medium category; correlation screen with one breach.
a03 OfferSearch_C	SR	Semantic offer search; Low category; adversarial text control.
a04 StressTest_D	stress	No consumer impact; Out of Scope.
a05 VendorScore_E	AF	Vendor fraud score; High category; DIA infeasible; vendor evidence package.
a06 PendingOffer_F	SR	Materiality unresolved; text inputs; no category recorded.
a07 ServicingText_G	SV	Servicing text model; Low category; redaction control.

B.3.2 Propositions, predicates, variables and constants

In propositional logic (PL) each symbol below is a statement about one assessment. In predicate logic (FOL) the variables are x (assessment), u (use), c, d, e (category), f (feature), p (protected-class contrast), r (correlation coefficient), t (text control) and k (checklist item). The constants are the assessments a01-a07; the uses AL (application lending), AF (application-stage fraud), DM (direct marketing), TF (transaction-stage fraud), SV (servicing), IM (indirect marketing), DP (deposits), SR (semantic offer search) and stress (stress testing); the categories H, M, L and O (Out of Scope for this review); the values yes/no and resolved/unresolved; the features income and utilisation; the contrasts race, age and gender; and the controls adversarial and redaction. The contrast labels stand for an approved encoding protocol, not arbitrary numeric coding of categories. Logical symbols: $\wedge$ and, $\vee$ or, $\neg$ not, $\rightarrow$ implies, $\leftrightarrow$ defined as, $\forall$ for all, $\exists$ there exists, $\exists!$ exactly one.

Propositions (PL). Input propositions are recorded facts; derived propositions are conclusions.

Symbol	Meaning	Symbol	Meaning
M	Registered assessment	ImpY, ImpN	Consumer impact recorded as yes / no
Res, Unres	Materiality resolved / unresolved	AL, AF, DM, TF, SV, IM, DP, SR	The use is that use
Mat	Material direct-marketing impact	AO, NM, SA	All offers accessible; non-material placement; servicing assessed
Tab, Txt	Tabular / text inputs	AttY, AttN	Protected attributes accessible / not accessible
FeY, FeN	DIA feasible / not feasible	ChH, ChM, ChL, ChO	Category selected by the analyst
Pack	Decision, rationale and legal guidance recorded	OvOK	Override rationale and acceptance recorded
Inv	Comparison inventory attested complete	CAacc, DIAacc, ALTacc	Analyst acceptance of correlation results, DIA interpretation, alternative route
Sel, AllDA	A text control is selected; every selected control is done and accepted	Vend, Aff, VL, Pil, SP	Vendor model; vendor affirmation; legal acceptance; pilot evidence; score-performance evidence
Req, Miss	Checklist item required; no valid record for it	Invalid	Input fails an integrity check
CC, FR, LR	Consumer-credit use; fair-lending review and legal review required	SgH, SgM, SgL	Suggested category (SoH, SoM, SoL: the only suggestion)
OvReq, Ovl	Override required; conflicting suggestions	FnH, FnM, FnL, FnO	Validated category
IS, CA, TXREQ, TXTOK	Input screening, correlation analysis, text-control decision required; text controls accepted	CADONE, CALOW, CABR, CAOK	Correlations complete; all below threshold; a breach exists; screen accepted

Symbol	Meaning	Symbol	Meaning
DIA, DIAREC, ALT, ALTOK	DIA required and recorded; alternative review required and satisfied	VREQ, VE, AR, Rej, Hold, Ready	Vendor review required; vendor evidence complete; attention required; rejected; blocked; ready

Predicates (FOL). Unary predicates take x; the others are listed with their arguments.

Predicate	Meaning
M(x), Use(x,u), UseClass(u,k), ConsumerCredit(x)	Registered assessment; its use; domain taxonomy of uses; consumer-credit use established
Impact(x,v), Materiality(x,v), Material(x), AllOffers(x), NonMaterial(x), ServicingAssessed(x)	Recorded impact and materiality values; material direct-marketing impact; offer-search conditions
Choice(x,c), Suggested(x,c), Final(x,c), CatOK(x,c)	Analyst choice; category suggested by the rules; validated category; choice aligned with the suggestion
Pack(x), OverrideOK(x), OverrideReq(x), Overlap(x)	Decision pack recorded; override accepted; override required; conflicting suggestions
Tabular(x), Text(x), Attr(x,v), Feas(x,v)	Input type; accessibility of protected attributes; DIA feasibility
Feature(x,f), Protected(x,p), Corr(x,f,p,r), NeedPair(x,f,p)	Declared feature and contrast; recorded coefficient; required comparison
InvComplete(x), CAAccept(x), CADONE(x), CALOW(x), CABREACH(x), CAOK(x)	Correlation screen: inventory, acceptance and the derived results
Selected(x,t), Done(x,t), Accepted(x,t), TXREQ(x), TXTOK(x)	Text-control selection, completion, acceptance and derived results
Vendor(x), Affirmation(x), VendorLegal(x), Pilot(x), ScorePerformance(x), VREQ(x), VE(x)	Vendor model, its evidence items and the derived requirement and evidence record
DIAAccept(x), AltAccept(x), DIA(x), DIAREC(x), ALT(x), ALTOK(x)	DIA and alternative-route acceptance, requirement and record
FR(x), LR(x), IS(x), CA(x), AR(x), Reject(x), Hold(x), Ready(x)	Review, legal review, screening and correlation analysis required; attention; rejection; hold; ready
Required(x,k), Missing(x,k), Invalid(x)	Checklist item required; intake audit found no valid record; integrity failure

B.3.3 Business rules and logical forms

Each business rule is stated in words, then in PL (one assessment) and FOL (all assessments). In PL, X stands for a category H, M, L or O, so R4-R7 and R11 each stand for one rule per category; CatOKX is defined in D4 below.

Business rule	Equivalent propositional (PL) and predicate (FOL) forms
R1 Lending uses are classified through domain knowledge.	PL: (AL ∨ AF) → CC FOL: ∀x ∀u [Use(x,u) ∧ UseClass(u,consumer_credit) → ConsumerCredit(x)]
R2 Consumer impact requires fair-lending review.	PL: M ∧ ImpY → FR FOL: ∀x [M(x) ∧ Impact(x,yes) → FR(x)]
R3 Unresolved materiality requires legal review.	PL: M ∧ Unres → LR FOL: ∀x [M(x) ∧ Materiality(x,unresolved) → LR(x)]
R4 Resolved consumer-credit use suggests High.	PL: FR ∧ Res ∧ CC → SgH FOL: ∀x [FR(x) ∧ Materiality(x,resolved) ∧ ConsumerCredit(x) → Suggested(x,H)]
R5 Material direct marketing or transaction fraud suggests Medium.	PL: FR ∧ Res ∧ ((DM ∧ Mat) ∨ TF) → SgM FOL: ∀x [FR(x) ∧ Materiality(x,resolved) ∧ ((Use(x,DM) ∧ Material(x)) ∨ Use(x,TF)) → Suggested(x,M)]
R6 Servicing, indirect marketing or deposit use suggests Low.	PL: FR ∧ Res ∧ (SV ∨ IM ∨ DP) → SgL FOL: ∀x [FR(x) ∧ Materiality(x,resolved) ∧ (Use(x,SV) ∨ Use(x,IM) ∨ Use(x,DP)) → Suggested(x,L)]

Business rule	Equivalent propositional (PL) and predicate (FOL) forms
R7 Non-material semantic offer search suggests Low if all offers are accessible and servicing is assessed.	PL: FR ∧ Res ∧ SR ∧ AO ∧ NM ∧ SA → SgL FOL: ∀x [FR(x) ∧ Materiality(x,resolved) ∧ Use(x,SR) ∧ AllOffers(x) ∧ NonMaterial(x) ∧ ServicingAssessed(x) → Suggested(x,L)]
R8 Conflicting suggestions need resolution; an override cannot bypass this.	PL: (SgH ∧ SgM) ∨ (SgH ∧ SgL) ∨ (SgM ∧ SgL) → Ovl ∧ AR FOL: ∀x ∀c ∀d [Suggested(x,c) ∧ Suggested(x,d) ∧ c ≠ d → Overlap(x) ∧ AR(x)]
R9 Choice that differs from the single suggested category requires an override.	PL: ChX ∧ SoY → OvReq, for X ≠ Y FOL: ∀x ∀c ∀d [Choice(x,c) ∧ Suggested(x,d) ∧ ∀e (Suggested(x,e) → e = d) ∧ c ≠ d → OverrideReq(x)]
R10 Required override without recorded rationale and acceptance needs attention.	PL: OvReq ∧ ¬OvOK → AR FOL: ∀x [OverrideReq(x) ∧ ¬OverrideOK(x) → AR(x)]
R11 Category is validated only with resolved materiality, a complete decision pack, an aligned choice and no conflict.	PL: M ∧ Res ∧ ChX ∧ Pack ∧ CatOKX ∧ ¬Ovl → FnX FOL: ∀x ∀c [M(x) ∧ Materiality(x,resolved) ∧ Choice(x,c) ∧ Pack(x) ∧ CatOK(x,c) ∧ ¬Overlap(x) → Final(x,c)]
R12 Validated Low or Medium category requires input screening.	PL: (FnL ∨ FnM) → IS FOL: ∀x [(Final(x,L) ∨ Final(x,M)) → IS(x)]
R13 Screening of tabular inputs with accessible attributes requires correlation analysis.	PL: IS ∧ Tab ∧ AttY → CA FOL: ∀x [IS(x) ∧ Tabular(x) ∧ Attr(x,yes) → CA(x)]
R14 Text inputs under fair-lending review require a text-control decision.	PL: FR ∧ Txt → TXREQ FOL: ∀x [FR(x) ∧ Text(x) → TXREQ(x)]
R15 Text controls are satisfied when at least one is selected and every selected control is done and accepted.	PL: TXREQ ∧ Sel ∧ AllDA → TXTOK FOL: ∀x [TXREQ(x) ∧ ∃t Selected(x,t) ∧ ∀t (Selected(x,t) → Done(x,t) ∧ Accepted(x,t)) → TXTOK(x)]
R16 High category with feasible DIA requires the DIA route.	PL: FnH ∧ FeY → DIA FOL: ∀x [Final(x,H) ∧ Feas(x,yes) → DIA(x)]
R17 High category with infeasible DIA requires alternative review.	PL: FnH ∧ FeN → ALT FOL: ∀x [Final(x,H) ∧ Feas(x,no) → ALT(x)]
R18 Vendor model under review, with infeasible DIA, requires vendor assurance.	PL: FR ∧ Vend ∧ AttN ∧ FeN → VREQ FOL: ∀x [FR(x) ∧ Vendor(x) ∧ Attr(x,no) ∧ Feas(x,no) → VREQ(x)]
R19 Complete, legally reviewed vendor package establishes vendor evidence.	PL: VREQ ∧ Aff ∧ VL ∧ Pil ∧ SP → VE FOL: ∀x [VREQ(x) ∧ Affirmation(x) ∧ VendorLegal(x) ∧ Pilot(x) ∧ ScorePerformance(x) → VE(x)]
R20 Required DIA with accepted interpretation is recorded.	PL: DIA ∧ DIAacc → DIAREC FOL: ∀x [DIA(x) ∧ DIAAccept(x) → DIAREC(x)]
R21 Vendor evidence closes the alternative route only with explicit acceptance.	PL: ALT ∧ VE ∧ ALTacc → ALTOK FOL: ∀x [ALT(x) ∧ VE(x) ∧ AltAccept(x) → ALTOK(x)]
R22 The correlation screen is satisfied by complete, uniformly low and accepted results.	PL: CA ∧ CADONE ∧ CALOW ∧ CAacc → CAOK FOL: ∀x [CA(x) ∧ CADONE(x) ∧ CALOW(x) ∧ CAAccept(x) → CAOK(x)]
R23 Correlation breach requires analyst attention.	PL: CABR → AR FOL: ∀x [CABREACH(x) → AR(x)]
R24 Every required item with no valid record is a review gap.	PL: Req ∧ Miss → AR (for each item k) FOL: ∀x ∀k [Required(x,k) ∧ Missing(x,k) → AR(x)]
R25 Invalid or contradictory input is rejected before other inference.	PL: Invalid → Rej FOL: ∀x [Invalid(x) → Reject(x)]
R26 Attention, legal referral or rejection blocks closure.	PL: (AR ∨ LR ∨ Rej) → Hold FOL: ∀x [(AR(x) ∨ LR(x) ∨ Reject(x)) → Hold(x)]
R27 Validated category, no hold and every triggered requirement satisfied make the checklist ready for governance review.	PL: (FnH ∨ FnM ∨ FnL ∨ FnO) ∧ ¬Hold ∧ (CA → CAOK) ∧ (TXREQ → TXTOK) ∧ (DIA → DIAREC) ∧ (ALT → ALTOK) → Ready FOL: ∀x [∃c Final(x,c) ∧ ¬Hold(x) ∧ (CA(x) → CAOK(x)) ∧ (TXREQ(x) → TXTOK(x)) ∧ (DIA(x) → DIAREC(x)) ∧ (ALT(x) → ALTOK(x)) → Ready(x)]

B.3.4 Supporting definitions

Definition	Formal form
D1	NeedPair(x,f,p) ↔ Feature(x,f) ∧ Protected(x,p) CADONE(x) ↔ InvComplete(x) ∧ ∃f ∃p NeedPair(x,f,p) ∧ ∀f ∀p [NeedPair(x,f,p) → ∃!r Corr(x,f,p,r)]
D2	CALOW(x) ↔ CADONE(x) ∧ ∀f ∀p ∀r [(NeedPair(x,f,p) ∧ Corr(x,f,p,r)) → |r| < 0.30]
D3	CABREACH(x) ↔ ∃f ∃p ∃r [NeedPair(x,f,p) ∧ Corr(x,f,p,r) ∧ |r| ≥ 0.30]
D4	CatOK(x,c) ↔ ¬Overlap(x) ∧ [(Impact(x,yes) ∧ c ∈ {H,M,L} ∧ Suggested(x,c) ∧ ∀e (Suggested(x,e) → e = c)) ∨ (Impact(x,yes) ∧ c ∈ {H,M,L} ∧ ∃d Suggested(x,d) ∧ OverrideOK(x)) ∨ (Impact(x,no) ∧ c = O)]

Propositional grounding of D1-D3. For a fixed assessment with n declared pairs, let Ci mean that pair i has exactly one valid coefficient, Bi that |ri| < 0.30 and Gi that |ri| ≥ 0.30. Then CADONE = Inv ∧ C1 ∧ … ∧ Cn, CALOW = CADONE ∧ B1 ∧ … ∧ Bn and CABR = G1 ∨ … ∨ Gn. For a02 (2 features × 3 contrasts) n = 6. A breach can be established while another comparison is still missing.

Missing and invalid evidence. An absent fact is unknown, not false: a missing feasibility record establishes neither Feas(x,yes) nor Feas(x,no), so neither DIA nor ALT is inferred. The intake audit lists the required items and records Missing(x,k) where no valid record exists (R24). R25 rejects conflicting values for impact, materiality, feasibility, accessibility or category, invalid category labels, undeclared coefficient pairs and coefficients outside [-1, 1]; conflicting duplicate coefficients must be resolved at intake. Hold and Ready (R26, R27) are evaluated last, after all other conclusions, and every change to a source record triggers a fresh evaluation from the source data, so an earlier hold or ready result cannot persist by accident.

B.3.5 Knowledge base

KB = F ∪ R ∪ D. F holds 84 ground facts: 80 assessment facts and 4 taxonomy facts. R holds the 27 inference rules R1-R27 and D holds the four definitions D1-D4. Derived results (FR, Final, CADONE, CAOK, Hold, Ready and so on) are not counted as facts. A Python script (Appendix A) evaluates the same rules procedurally over the facts; it is a check on the examples, not a general inference engine.

Case	Explicit facts	Count
Taxonomy	UseClass(AL, consumer_credit); UseClass(AF, consumer_credit); UseClass(DM, marketing); UseClass(SV, servicing)	4
a01	M(a01); Use(a01, AL); Impact(a01, yes); Materiality(a01, resolved); Choice(a01, H); Pack(a01); Feas(a01, yes); DIAAccept(a01); Tabular(a01); Attr(a01, yes)	10
a02	M(a02); Use(a02, DM); Impact(a02, yes); Materiality(a02, resolved); Choice(a02, M); Pack(a02); Material(a02); Tabular(a02); Attr(a02, yes); Feature(a02, income); Feature(a02, utilisation); Protected(a02, age); Protected(a02, gender); Protected(a02, race); InvComplete(a02); CAAccept(a02); Corr(a02, income, age,-0.25); Corr(a02, income,gender,0.10); Corr(a02,income, race,0.12); Corr(a02, utilisation, age,0.34); Corr(a02, utilisation, gender,-0.15); Corr(a02, utilisation, race,0.08)	22
a03	M(a03); Use(a03, SR); Impact(a03, yes); Materiality(a03, resolved); Choice(a03, L); Pack(a03); AllOffers(a03); NonMaterial(a03); ServicingAssessed(a03); Text(a03); Selected(a03, adversarial); Done(a03, adversarial); Accepted(a03, adversarial)	13
a04	M(a04); Use(a04, stress); Impact(a04, no); Materiality(a04, resolved); Choice(a04, O); Pack(a04)	6
a05	M(a05); Use(a05,AF); Impact(a05,yes); Materiality(a05,resolved); Choice(a05,H); Pack(a05); Feas(a05,no); Attr(a05,no); Vendor(a05); Affirmation(a05); VendorLegal(a05); Pilot(a05); ScorePerformance(a05)	13
a06	M(a06); Use(a06,SR); Impact(a06,yes); Materiality(a06,unresolved); Text(a06); AllOffers(a06)	6
a07	M(a07); Use(a07, SV); Impact(a07, yes); Materiality(a07, resolved); Choice(a07, L); Pack(a07); Text(a07); Selected(a07, redaction); Done(a07, redaction); Accepted(a07, redaction)	10

B.3.6 Workplace queries

Each query states a goal, evaluates the recorded facts and lists the rules that support the answer. Where a query compares snapshots, one item changes at a time. There are five queries and 15 snapshots in total.

Query 1. Which coefficient prevents Marketing_B (a02) from closing, and does 0.30 pass?

Feature	Race	Age	Gender
income	0.12	-0.25	0.10
utilisation	0.08	**0.34**	-0.15

Goal: Hold(a02). **Answer: yes.** (1) R5 suggests Medium, the analyst chose Medium and the decision pack is recorded, so R11 validates Medium. (2) R12 requires input screening and R13 correlation analysis (tabular inputs, accessible attributes). (3) All six pairs are present and the inventory is attested, so D1 gives CADONE(a02). (4) Corr(a02,utilisation,age,0.34) satisfies D3, giving CABREACH(a02); R23 gives AR and R26 gives Hold. (5) D2 fails, so R22 cannot establish CAOK. The Medium category itself remains valid.

Snapshot (only this coefficient changed)	Threshold evaluation	Checklist outcome
Original +0.34	$\|0.34\| \geq 0.30$: breach	HOLD; CAOK not established
Replace with +0.29	Maximum $\|r\| = 0.29 < 0.30$	CALOW, CAOK; READY (R27)
Replace with +0.30	Equality breaches the strict limit	HOLD
Replace with -0.30	Absolute-value equality also breaches	HOLD

The analyst should investigate the utilisation-age result. The +0.29 row only tests boundary behaviour: CAAccept(a02) is carried over from the baseline for the test, whereas in practice a changed coefficient needs a documented basis and fresh acceptance. Passing this screen covers only the stated correlation requirement.

Query 2. Can an unexplained High-to-Low override be recorded for Underwrite_A (a01)?

Goal: Final(a01,L) after the analyst selects Low. **Answer: no, not without an accepted override.** R1 and R4 suggest High as the only category. Choosing Low triggers R9 (OverrideReq). Without a recorded rationale and acceptance, R10 gives AR and R26 gives Hold. CatOK(a01,L) in D4 also needs OverrideOK, so R11 cannot validate Low.

Snapshot	Category outcome	Outstanding action
Low selected; no override record	No validated category; HOLD	Record a separate override rationale and authorised acceptance.
Override rationale and acceptance recorded	Final(a01,L) by R11; HOLD remains	Low triggers R12 and R13 (tabular, accessible attributes), but no features, coefficients or CAAccept are recorded; R24 raises the gaps.

Overlapping suggestions (R8) are not exercised by these snapshots.

Query 3. What text-control action remains for each text case?

Goal: find the remaining actions for $FR(x) \wedge Text(x)$. The matching cases are a03, a06 and a07.

Case	Evidence and rule links	Outcome and remaining action
a03 OfferSearch_C	R7 suggests Low (semantic search, all offers accessible, non-material, servicing assessed); R11 validates Low. R14 requires a text-control decision. Adversarial control is selected, done and accepted, so R15 gives TXTOK.	TXTOK; READY. No outstanding text control.
a06 PendingOffer_F	R3 requires legal review because materiality is unresolved. R14 requires a text-control decision, but no control is selected. No category or decision pack is recorded.	LR, AR, HOLD. Resolve materiality; record category, rationale and legal guidance; select, complete and accept the applicable controls.
a07 ServicingText_G	R6 suggests Low from servicing; R11 validates Low. R14 requires a text-control decision. Redaction is selected, done and accepted, so R15 gives TXTOK.	TXTOK; READY. No outstanding text control.

R15 checks every selected control, so a second selected control would also need completion and acceptance. The snapshots cover single-control cases only, and an analyst must still judge whether the selected controls suit the model.

Query 4. Does missing or contradictory evidence block the review?

Goal: Hold(x) or Reject(x) after an evidence change. Each row starts from its own baseline.

Isolated test	Inference or integrity result	Action
a01: remove Feas(a01,yes)	High remains validated; neither DIA nor ALT is inferred because feasibility is unknown. The audit records the gap, so AR and Hold follow (R24, R26).	Obtain a feasibility assessment; do not infer infeasibility.
a04: remove Impact(a04,no)	No-impact evidence is unknown, so CatOK(a04,O) fails and Out of Scope cannot be validated. Gaps in impact and category alignment give Hold.	Obtain explicit impact evidence.
a02: remove the income-race coefficient	D1 fails, so CADONE and CALOW cannot be established. The +0.34 value still gives CABREACH.	Complete the missing comparison and investigate the breach.
a04: record Impact yes and no	R25 rejects the input before any category is derived; R26 holds it.	Resolve the conflicting source records, then rebuild the snapshot.

Query 5. What additional decision closes the vendor alternative route for VendorScore_E (a05)?

Goal: ALTOK(a05). **Answer: not for the baseline record.** R1 and R4 suggest High and R11 validates it. Recorded DIA infeasibility triggers ALT (R17). Vendor opacity triggers VREQ (R18) and the four package records establish VE (R19). R21 still needs ALTacc.

Snapshot	Supported result	Closure decision
a05 baseline	ALT, VREQ and VE are established; AltAccept is absent.	ALTOK not established; R24 gives AR and HOLD.
Add AltAccept(a05)	ALT ∧ VE ∧ AltAccept establishes ALTOK (R21).	No other gaps remain: READY by R27.

The acceptance is an accountable human decision on whether the vendor package is adequate for the purpose and limits of the review.

Release of holds and scope of Ready. A hold is released only by a new snapshot. For a06 this needs resolved materiality, a valid category and complete evidence; the displayed runs show only the held baseline. Ready means the checklist is complete for governance review. Deployment approval, legal advice and wider fairness assessment stay with the authorised reviewers, and Out of Scope applies to this fair-lending review only.

B.4 Analyse (1 mark)

Expressiveness of predicate logic compared with propositional logic

Workplace need	Propositional representation	Predicate representation
Check every declared comparison	Six coefficient-present symbols and six comparison results for a02; each new feature or contrast needs new symbols.	∀f ∀p ranges over declared pairs; D1 checks coverage and D2 checks all values.
Identify the specific breach	An aggregate breach flag hides which comparison failed unless extra ground symbols are added.	∃f ∃p ∃r in D3 gives the witness: utilisation and age at +0.34.
Compare chosen and suggested categories	One rule for each pair of categories, repeated for each case.	Choice(x,c), Suggested(x,d) and c ≠ d state the relationship once (R9).
Require every selected text control	One conjunction for each possible selection.	∀t [Selected(x,t) → Done(x,t) ∧ Accepted(x,t)] handles any selection (R15).
Reuse domain knowledge	The use-classification rule is repeated for each assessment.	Use(x,u) joined with UseClass(u,consumer_credit) lets new cases reuse R1.

Predicate logic lets one rule cover all assessments, features and controls, and lets a query return the specific item that caused a result, such as the utilisation-age pair. A finite propositional version can encode the same decisions but needs many more grounded symbols and a new rule set whenever a case or feature is added. Neither form removes the need for an approved arithmetic check and an approved contrast-encoding protocol.

Effectiveness of the knowledge base

Checklist area	Rules	Coverage finding
Business-use classification	R1, R25	Use and taxonomy represented; upstream decision on whether a tool is a model is excluded.
Impact and materiality	R2-R3, R24-R26	Unknown values create gaps; unresolved materiality creates a legal referral.
Suggested and validated category	R4-R11, D4	Overrides and overlaps are explicit; the validated category is unique.
Tabular input screening	R12-R13, R22-R23, D1-D3	Threshold, completeness and acceptance are separate conditions.
Text controls	R14-R15	Every selected control must be completed and accepted.
High-risk DIA route	R16-R17, R20	Feasibility is required; accepted interpretation is separate from the recommendation.
Vendor alternative evidence	R18-R21	Receiving the package does not close the route without acceptance.
Closure	R24-R27	Gaps and contradictions block closure; a new complete snapshot can release a hold.

The 15 snapshots give the expected outcome in the script, so the rules behave as intended on the examples. This does not show coverage of the full workplace policy or performance on real review records. For seven fixed cases a decision table would be easier to maintain; the knowledge base pays off as the inventory grows and the same relationships and evidence checks recur. It also makes two points hard to miss: in a02 analyst acceptance cannot clear a recorded breach, and in a05 receiving the vendor package does not close the alternative route.

Missing knowledge and limitations

The knowledge base does not cover sample suitability, statistical power, evidence expiry, reviewer authority or wider fairness assessment, which remain with the analyst. The 0.30 threshold is an assumption: no approved policy extract or calibration is available here. Race and gender labels need an approved contrast and encoding protocol before a correlation can be interpreted. Low or Medium tabular cases with inaccessible attributes, and non-vendor alternative routes, are held for route resolution rather than resolved. The snapshots do not test overlapping suggestions, transaction fraud, indirect marketing or deposits. Before use, an analyst and the legal reviewer should check the use classifications, override authority, evidence requirements and hold reasons against approved policy and past reviews.

B.5 Reflect (1 mark)

Constraint	Effect in my workplace
Incomplete or stale records	A missing feasibility record must not be treated as infeasibility, and a stale coefficient could wrongly keep a case ready. Facts need a source, a date and a review period.
Evolving policy	Use classifications, thresholds and acceptable alternatives need versioning with effective dates, and regression tests when they change.
Privacy and access	Protected-class data and customer data need an approved purpose, restricted access and retention limits. This assignment uses synthetic records only.
Vendor opacity and judgement	A vendor package may lack independence or population coverage, so accepting an alternative route must stay an accountable human decision.
Technology and scale	Many features and contrasts produce many pairs, so evaluation needs indexing and completeness checks. Re-evaluating from source data avoids stale conclusions but costs processing time.
Input integrity	The ingest layer must keep source conflicts and reject conflicting duplicate coefficients before building the keyed records.

Future scope of improvement

I would first build an evidence-intake gate tied to the policy version and the source records. The risk analyst would own the required-item list and the acceptance records, the legal reviewer would review exceptions, and the model owner would supply the evidence. A pilot on the seven cases here could record missing items and source conflicts and agree freshness limits before wider use.

I recommend this because the examples show that receiving evidence, completing coverage, accepting the evidence and closing the review are separate decisions. A missing feasibility record or a conflicting impact entry can block an otherwise complete case, and a rule base is only as reliable as the records fed into it. The gate should catch those gaps at entry and trigger a fresh evaluation whenever a source record changes.

Appendix A - Python Implementation

The script below evaluates R1-R27 and D1-D3 for the seven cases and reproduces every snapshot in Queries 1-5.

```
"""Fair-lending validation checklist: rule-based KB.
Run: python Fair_Lending_KB.py """

from copy import deepcopy
from decimal import Decimal

LIMIT = Decimal('0.30')
TAXONOMY = {'AL': 'consumer_credit', 'AF': 'consumer_credit',
            'DM': 'marketing', 'SV': 'servicing'}

def case(use, choice, **kw):
    d = dict(use={use}, impact={'yes'}, mat={'resolved'}, choice={choice}, pack=True)
    d.update(kw)
    return d

CASES = {
 'a01': case('AL', 'H', feas={'yes'}, dia_ok=True, tab=True, attr={'yes'}),
 'a02': case('DM', 'M', material=True, tab=True, attr={'yes'},
             feat={'income', 'utilisation'}, prot={'race', 'age', 'gender'},
             inv=True, ca_ok=True,
             corr={('income','race'): '0.12', ('income','age'): '-0.25',
                   ('income','gender'): '0.10', ('utilisation','race'): '0.08',
                   ('utilisation','age'): '0.34', ('utilisation','gender'): '-0.15'}),
 'a03': case('SR', 'L', all_offers=True, nonmat=True, svc=True, text=True,
             sel={'adversarial'}, done={'adversarial'}, acc={'adversarial'}),
 'a04': case('stress', 'O', impact={'no'}),
 'a05': case('AF', 'H', feas={'no'}, attr={'no'}, vendor=True, aff=True,
             vlegal=True, pilot=True, score=True),
 'a06': dict(use={'SR'}, impact={'yes'}, mat={'unresolved'}, text=True, all_offers=True),
 'a07': case('SV', 'L', text=True, sel={'redaction'}, done={'redaction'}, acc={'redaction'}),
}

def evaluate(d):
    out, miss, bad, sug, why = set(), set(), set(), set(), {}
    def add(n, r): out.add(n); why.setdefault(n, []).append(r)
    for k, ok in [('impact', {'yes','no'}), ('mat', {'resolved','unresolved'}),
                  ('feas', {'yes','no'}), ('attr', {'yes','no'}), ('choice', set('HMLO'))]:
        v = d.get(k, set())
        if len(v) > 1 or not v <= ok: bad.add(k)
    pairs = {(f, p) for f in d.get('feat', ()) for p in d.get('prot', ())}
    for q, r in d.get('corr', {}).items():
        if abs(Decimal(r)) > 1 or q not in pairs: bad.add('corr')
    if bad:                                      # R25, R26
        add('REJECT', 'R25'); add('HOLD', 'R26')
        return out, miss, bad, sug, why
    imp, res = d.get('impact') == {'yes'}, d.get('mat') == {'resolved'}
    for k in ('impact', 'mat', 'choice'):
        if not d.get(k): miss.add(k)
    if not d.get('pack'): miss.add('pack')
    if imp: add('FR', 'R2')
    if d.get('mat') == {'unresolved'}: add('LR', 'R3')
    u = d['use']
    if any(TAXONOMY.get(x) == 'consumer_credit' for x in u): add('CC', 'R1')
    if imp and res:
        if 'CC' in out: sug.add('H'); add('SUG_H', 'R4')
        if ('DM' in u and d.get('material')) or 'TF' in u: sug.add('M'); add('SUG_M', 'R5')
        if u & {'SV', 'IM', 'DP'}: sug.add('L'); add('SUG_L', 'R6')
        if 'SR' in u and all(d.get(k) for k in ('all_offers', 'nonmat', 'svc')):
            sug.add('L'); add('SUG_L', 'R7')
    if len(sug) > 1: add('OVERLAP', 'R8'); add('AR', 'R8')
    c = next(iter(d.get('choice', ())), None)
    ovr_ok = d.get('ovr_ok')
    if c and len(sug) == 1 and c not in sug:            # R9, R10
        add('OVR_REQ', 'R9')
        if not ovr_ok: miss.add('override'); add('AR', 'R10')
    catok = (imp and c in ('H','M','L') and 'OVERLAP' not in out and
             (sug == {c} or (sug and ovr_ok))) or (d.get('impact') == {'no'} and c == 'O')
    if not catok: miss.add('category_alignment')
    if res and d.get('pack') and catok: add('FINAL_' + c, 'R11')
    need = set()
    if {'FINAL_L', 'FINAL_M'} & out: add('IS', 'R12')
    if 'IS' in out:
        if not (d.get('tab') or d.get('text')): miss.add('input_route')
        if d.get('tab'):
            if d.get('attr') == {'yes'}: add('CA', 'R13'); need.add('CAOK')
            else: miss.add('accessible_tabular_route')
    if imp and d.get('text'):                           # R14, R15
        add('TXREQ', 'R14'); need.add('TXTOK')
        sel = d.get('sel', set())
        if not sel: miss.add('text_selection')
        for t in sel:
            if t not in d.get('done', ()) or t not in d.get('acc', ()): miss.add('text_' + t)
```

```
        if sel and sel <= d.get('done', set()) and sel <= d.get('acc', set()): add('TXTOK', 'R15')
    if 'FINAL_H' in out:
        if not d.get('feas'): miss.add('feasibility')
        if d.get('feas') == {'yes'}:                    # R16, R20
            add('DIA', 'R16'); need.add('DIAREC')
            if d.get('dia_ok'): add('DIAREC', 'R20')
            else: miss.add('dia_accept')
        if d.get('feas') == {'no'}: add('ALT', 'R17'); need.add('ALTOK')
    if imp and d.get('vendor') and d.get('attr') == {'no'} and d.get('feas') == {'no'}:
        add('VREQ', 'R18')                              # R18, R19
        pk = [d.get(k) for k in ('aff', 'vlegal', 'pilot', 'score')]
        if all(pk): add('VE', 'R19')
        else: miss.add('vendor_package')
    if 'ALT' in out:
        if 'VE' not in out: miss.add('alternative_package')
        if not d.get('alt_ok'): miss.add('alternative_accept')
        elif 'VE' in out: add('ALTOK', 'R21')
    done = bool(pairs) and d.get('inv') and pairs <= d.get('corr', {}).keys()
    if 'CA' in out:
        if not done: miss.add('correlation_coverage')
        if not d.get('ca_ok'): miss.add('ca_accept')
    if done: add('CADONE', 'D1')
    if done and all(abs(Decimal(d['corr'][q])) < LIMIT for q in pairs): add('CALOW', 'D2')
    if any(abs(Decimal(r)) >= LIMIT for r in d.get('corr', {}).values()):
        add('CABREACH', 'D3'); add('AR', 'R23')
    if {'CADONE', 'CALOW'} <= out and 'CA' in out and d.get('ca_ok'): add('CAOK', 'R22')
    if miss: add('AR', 'R24')
    if out & {'AR', 'LR', 'REJECT'}: add('HOLD', 'R26')
    if any(x.startswith('FINAL_') for x in out) and need <= out and 'HOLD' not in out:
        add('READY', 'R27')
    return out, miss, bad, sug, why

RULE = {
 'R1': 'the recorded use is classed as consumer credit',
 'R2': 'recorded consumer impact requires fair-lending review',
 'R3': 'unresolved materiality requires legal review',
 'R4': 'resolved consumer-credit use suggests High',
 'R5': 'material direct marketing (or transaction fraud) suggests Medium',
 'R6': 'servicing, indirect marketing or deposits suggest Low',
 'R7': 'non-material semantic offer search suggests Low',
 'R8': 'conflicting suggestions need resolution',
 'R9': 'the choice differs from the single suggestion, so an override is required',
 'R10': 'override rationale or acceptance is missing',
 'R11': 'resolved materiality, decision pack and aligned choice validate the category',
 'R12': 'Low or Medium requires input screening',
 'R13': 'tabular inputs with accessible attributes need correlation analysis',
 'R14': 'text inputs under review need a text-control decision',
 'R15': 'every selected text control is done and accepted',
 'R16': 'High with feasible DIA requires the DIA route',
 'R17': 'High with infeasible DIA requires alternative review',
 'R18': 'opaque vendor model with infeasible DIA requires vendor assurance',
 'R19': 'the vendor package is complete',
 'R20': 'DIA interpretation is accepted and recorded',
 'R21': 'vendor evidence plus explicit acceptance closes the alternative route',
 'R22': 'complete, all-below-threshold and accepted correlations satisfy the screen',
 'R23': 'a correlation breach needs analyst attention',
 'R24': 'a required record is missing',
 'R25': 'invalid or contradictory input is rejected',
 'R26': 'attention, legal referral or rejection blocks closure',
 'R27': 'validated category, no hold and all obligations met: ready for governance review',
 'D1': 'all declared feature/contrast pairs have one coefficient and the inventory is complete',
 'D2': 'every coefficient satisfies |r| < 0.30',
 'D3': 'at least one coefficient has |r| >= 0.30',
}
NAME = {'H': 'High', 'M': 'Medium', 'L': 'Low', 'O': 'Out of Scope'}

def run(label, d):
    """Evaluate one snapshot and print a text answer with its reasoning."""
    out, miss, bad, sug, why = evaluate(d)
    fin = sorted(NAME[x[6:]] for x in out if x.startswith('FINAL_'))
    status = 'REJECTED' if 'REJECT' in out else 'READY' if 'READY' in out else 'HOLD'
    print(f"  [{label}] ANSWER: {status}. Validated category: {', '.join(fin) or 'not established'}.")
    print('    REASONING:')
    n = 0
    for rule in sorted({r for rs in why.values() for r in rs}, key=lambda r: (r[0], int(r[1:]))):
        got = ', '.join(sorted(k for k, rs in why.items() if rule in rs))
        n += 1; print(f"    {n}. {rule}: {RULE[rule]} -> {got}")
    if miss:
        n += 1; print(f"    {n}. Intake audit found no valid record for: {', '.join(sorted(miss))} (unknown, not false).")
    if bad:
        n += 1; print(f"    {n}. Integrity check failed for: {', '.join(sorted(bad))}; evaluate a fresh snapshot.")
    return status

def query(n, text):
    print(f"\nQUERY {n}: {text}")

if __name__ == '__main__':
```

```
query(1, 'Which coefficient prevents Marketing_B (a02) from closing, and does 0.30 pass?')
base = run('baseline', CASES['a02'])
res = {}
for v in ('0.29', '0.30', '-0.30'):
    d = deepcopy(CASES['a02']); d['corr'][('utilisation', 'age')] = v
    res[v] = run('utilisation-age = ' + v, d)
print(f"  QUERY ANSWER: utilisation/age (r = 0.34) breaches the 0.30 limit, so a02 is on {base}. "
      f"At 0.29 the result is {res['0.29']}; at +0.30 and -0.30 it is {res['0.30']} because the limit is strict on |r|.")

query(2, 'Can an unexplained High-to-Low override be recorded for Underwrite_A (a01)?')
d = deepcopy(CASES['a01']); d['choice'] = {'L'}
r1 = run('Low selected, no override', d)
d['ovr_ok'] = True
r2 = run('Low selected, override accepted', d)
print(f"  QUERY ANSWER: No. Without an accepted override the category is not validated ({r1}). "
      f"With one, Low is validated but the checklist stays on {r2} until the correlation evidence is supplied.")

query(3, 'What text-control action remains for each text case (a03, a06, a07)?')
r = {x: run(x, CASES[x]) for x in ('a03', 'a06', 'a07')}
print(f"  QUERY ANSWER: a03 {r['a03']} and a07 {r['a07']} need no further text-control action. "
      f"a06 is on {r['a06']}: resolve materiality, record a category and decision pack, then select, complete and accept a control.")

query(4, 'Does missing or contradictory evidence block the review?')
d = deepcopy(CASES['a01']); del d['feas']; s1 = run('a01 without feasibility', d)
d = deepcopy(CASES['a04']); del d['impact']; s2 = run('a04 without impact', d)
d = deepcopy(CASES['a02']); del d['corr'][('income', 'race')]; s3 = run('a02 without income-race coefficient', d)
d = deepcopy(CASES['a04']); d['impact'] = {'yes', 'no'}; s4 = run('a04 with impact yes and no', d)
print(f"  QUERY ANSWER: Yes. Missing evidence gives {s1}, {s2} and {s3}; contradictory evidence is {s4} before any category is derived.")

query(5, 'What additional decision closes the vendor alternative route for VendorScore_E (a05)?')
b = run('baseline', CASES['a05'])
d = deepcopy(CASES['a05']); d['alt_ok'] = True
a = run('alternative accepted', d)
print(f"  QUERY ANSWER: An explicit alternative-review acceptance (AltAccept). Without it a05 is on {b}; with it a05 is {a}.")
```

Appendix B - Output Screenshot

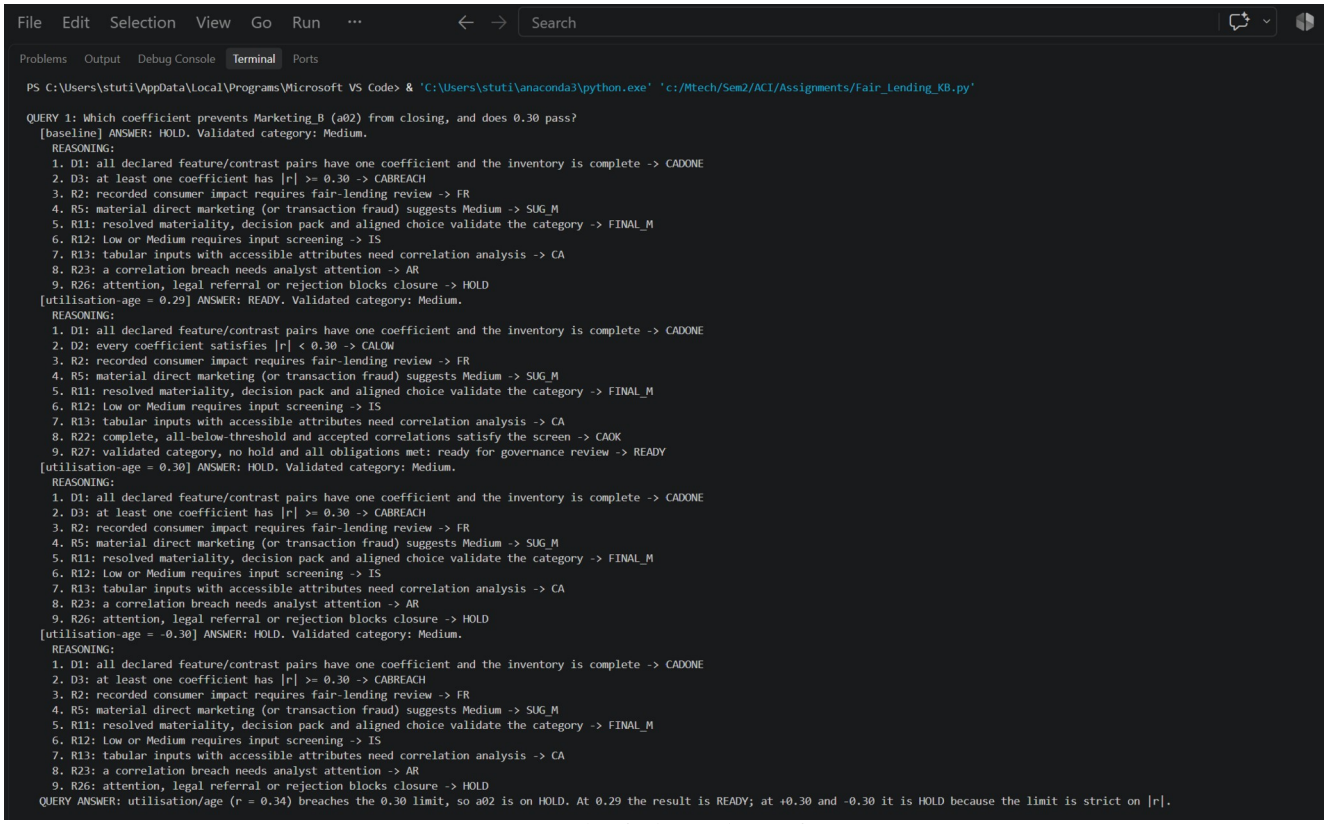


Figure 1: Snapshot of code run and output of Query 1

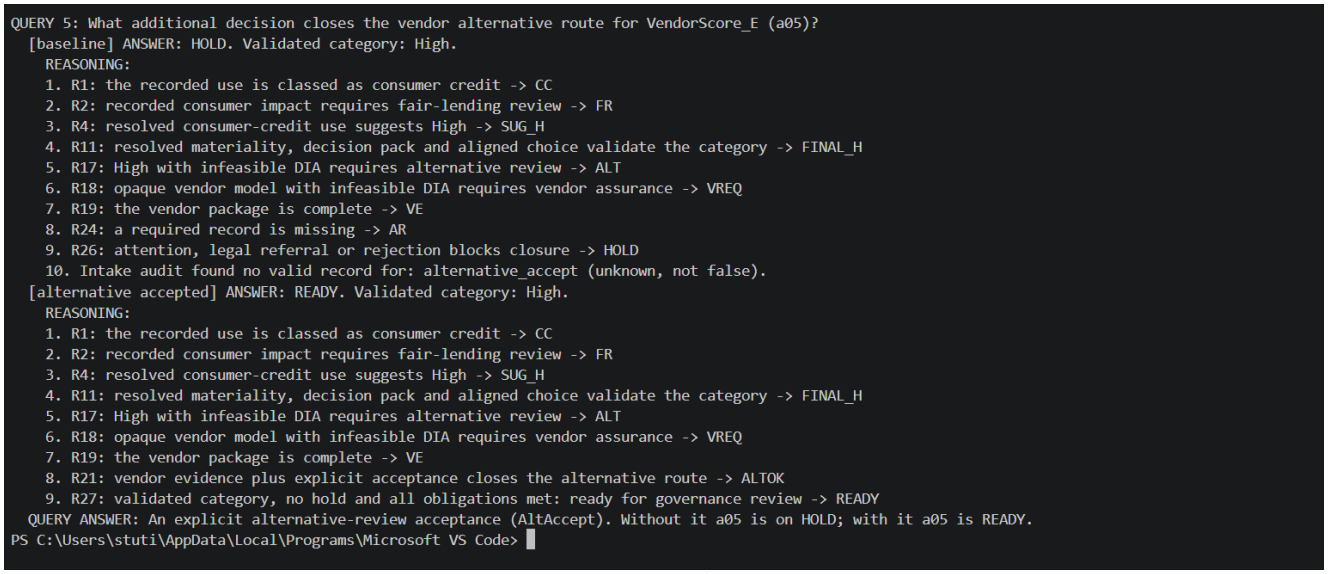


Figure 2: Output of Query 5